

Evaluating Future Flood Risks in Rockaway, NY through Geomorphological Analysis & Flood Simulations

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Abstract:

Coastal flooding is becoming a more severe issue due to rising sea levels caused by climate change. Long Island, New York, is particularly vulnerable to high rates of flooding because of its topography. Rockaway, located in the Queens borough of New York City, is only about 1.5 meters above sea level, which makes it extremely vulnerable to flooding from both the Atlantic Ocean to the south and Jamaica Bay to the north. This vulnerability is exacerbated by its flat topography and low elevation. In order to illustrate the high risk of flooding, researchers applied Geographical Information Systems (GIS) methods to simulate and assess future flood rates in Rockaway. By using GIS software Global Mapper and Light Detection and Ranging (LiDAR) data, a high-resolution digital elevation model (DEM) was constructed. This bare-earth model excluded trees, buildings, and other surface objects to enable flood simulations and provide a more complete understanding of the flood risks associated with different geomorphological regions of Long Island, specifically those of Rockaway. The simulations showed that most of Rockaway has a very low elevation, especially in the southwestern parts of the island, and these areas should be a first priority for first responders due to the high flooding risk. Areas in the middle of Rockaway have a lower risk of flooding and should be addressed afterwards. The data collected from this study can be useful in many different regions of the world that may be affected by rising sea levels or areas that are more prone to flooding. This information can help prepare first responders and provide them with a map of areas that are most at risk. Furthermore, the study highlights the importance of using GIS and LiDAR techniques to construct accurate models for flood simulations. By doing so, communities can better prepare for potential disasters and minimize the damage caused by flooding. Overall, this study serves as a valuable resource for anyone interested in understanding the impact of rising sea levels on vulnerable coastal communities and how to effectively manage flood risks.

Background Information:

- By 2100, global sea level is projected to increase by 0.5 to 1 meter (Horton et al. 2014)
- Rockaway, NY is approximately 1.5 meters above sea level with a population of 125,000 as of 2020.
- The rising sea levels are connected to the melting of glaciers and ice sheets as well as the thermal expansion of the ocean as the water warms.
- The Rockaway Peninsula frequently floods due to saltwater coming up from sewers and rain that will not drain through sewers blocked by high sea levels.
- Waterfront Alliance estimated that 61% of Rockaway residents have a one-in-two chance of a major flood in their home by 2060.
- Hurricane Sandy destroyed more than 1,000 homes on the Rockaway Peninsula due to the 10-foot storm surge flooding the area.
- Ten years later, New York State has spent \$11 billion to repair the damages caused by the storm and some areas still haven't completely recovered.

Method/Materials:

- Used Google Earth to create a shapefile of Rockaway
- Retrieved LiDAR data from the New York state government website
 - Built a digital elevation model of Rockaway
- Created a bare earth model by isolating ground return LiDAR points
 - A grid was applied to this area with cells of 500 m by 500 m within each cell a flood simulation was created
- Values of simulated elevation on a flat polygon (simulated flood) ranged from -8 to 10 meters
 - Data was taken every 0.1 meters and the first derivative of the simulation data was taken to observe the slope of the volumetric-fill and that of the area-fill at a given elevation
 - The cells with the lowest and highest data were standardized and compared (Fig. 6)

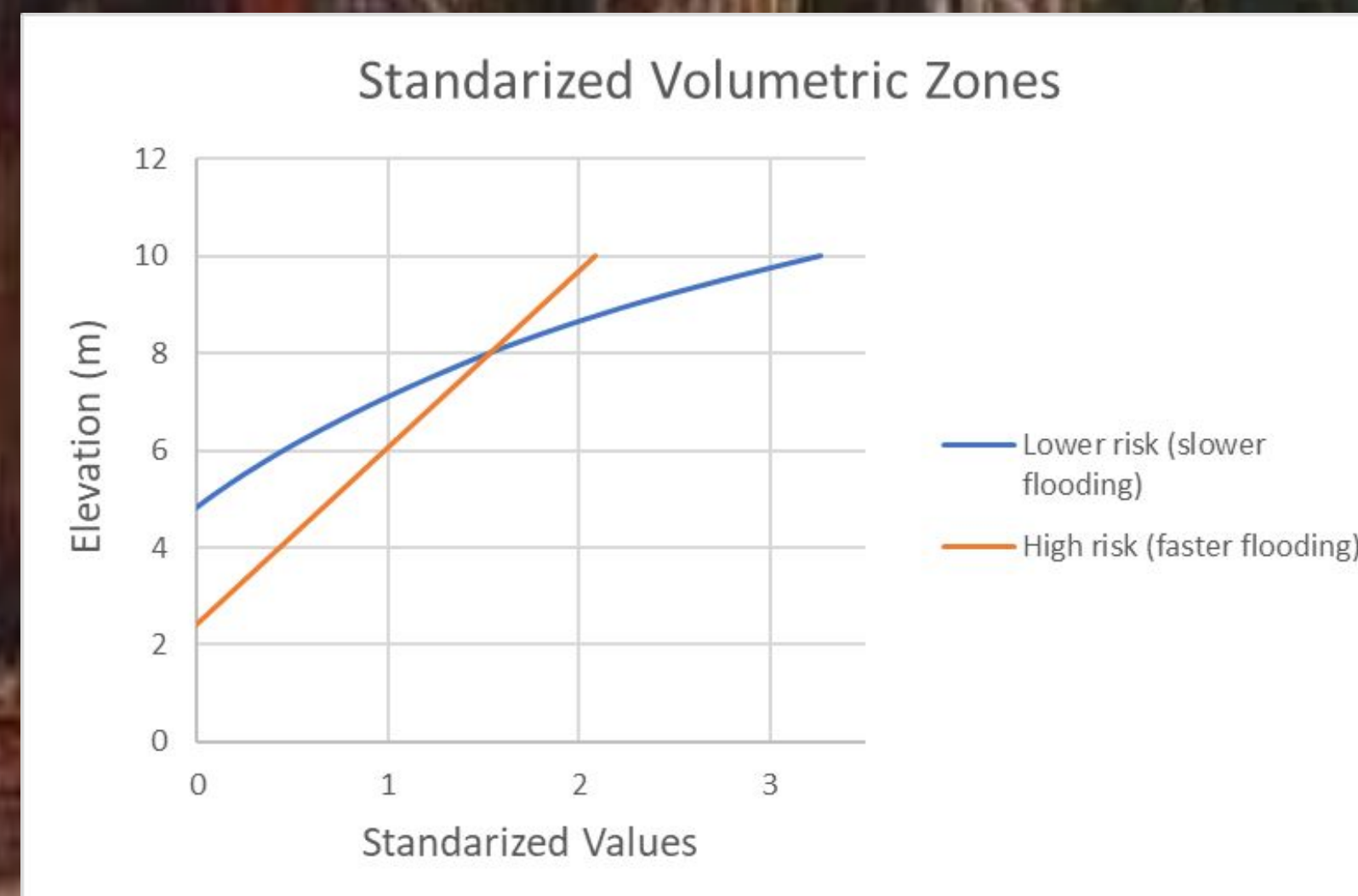


Figure 1: Standardized values of the lowest and highest flood rate volumetric cells to see the flood rate windows of the cells.



Figure 2: Standardized values of the lowest and highest cells of flood rate by area to see the flood rate windows of the cells.

Results:

- The classification maps were sorted into four different risk levels; low, moderate, high, and very high (Fig. 3 and 4)
- Volumetric map shows a majority of high risk areas in the southwestern side of the island while the central and eastern parts are mainly a low to moderate risk (Fig. 3)
- The classification map for the area doesn't show as great of contrast as a majority of regions show a high risk of flooding (Fig. 4)
- The lowest and highest flood rate cell per volume and area was then graphed with the slopes being 18007.58 m³, 164026.3 m³, 0.0059m², and 0.0205m² per 0.1m change in elevation respectively.
- The lowest and highest cells data were then standardized and the flood rate window is much smaller in the high risk areas compared to the low risk which have a bigger window (Fig. 1 and 2)

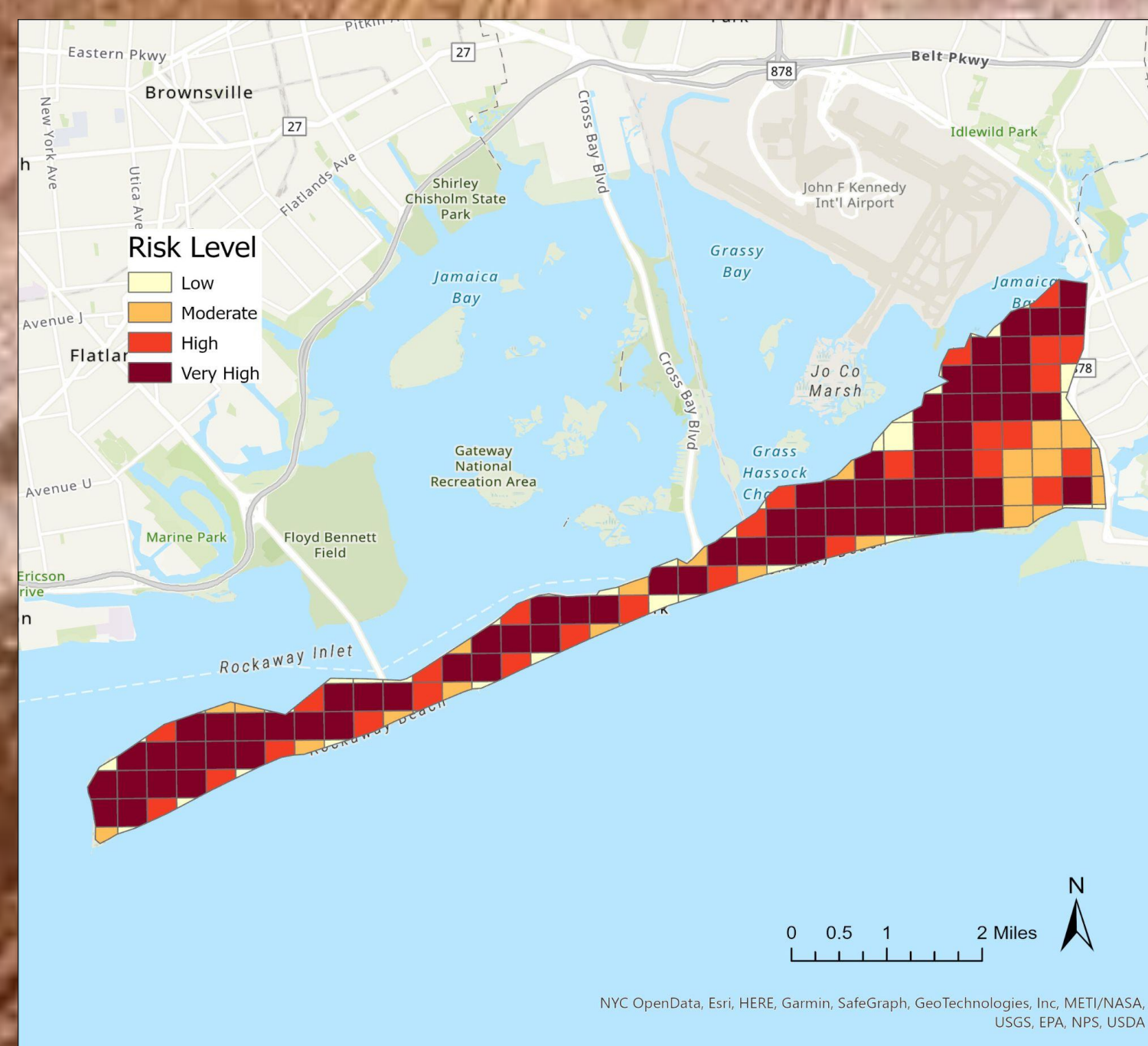


Figure 3: A classification GIS map that displays the risk level per cell of the grid based on flood rate per area. The risk levels were categorized into 4 levels ranging from low (yellow color) to very high (dark red color).

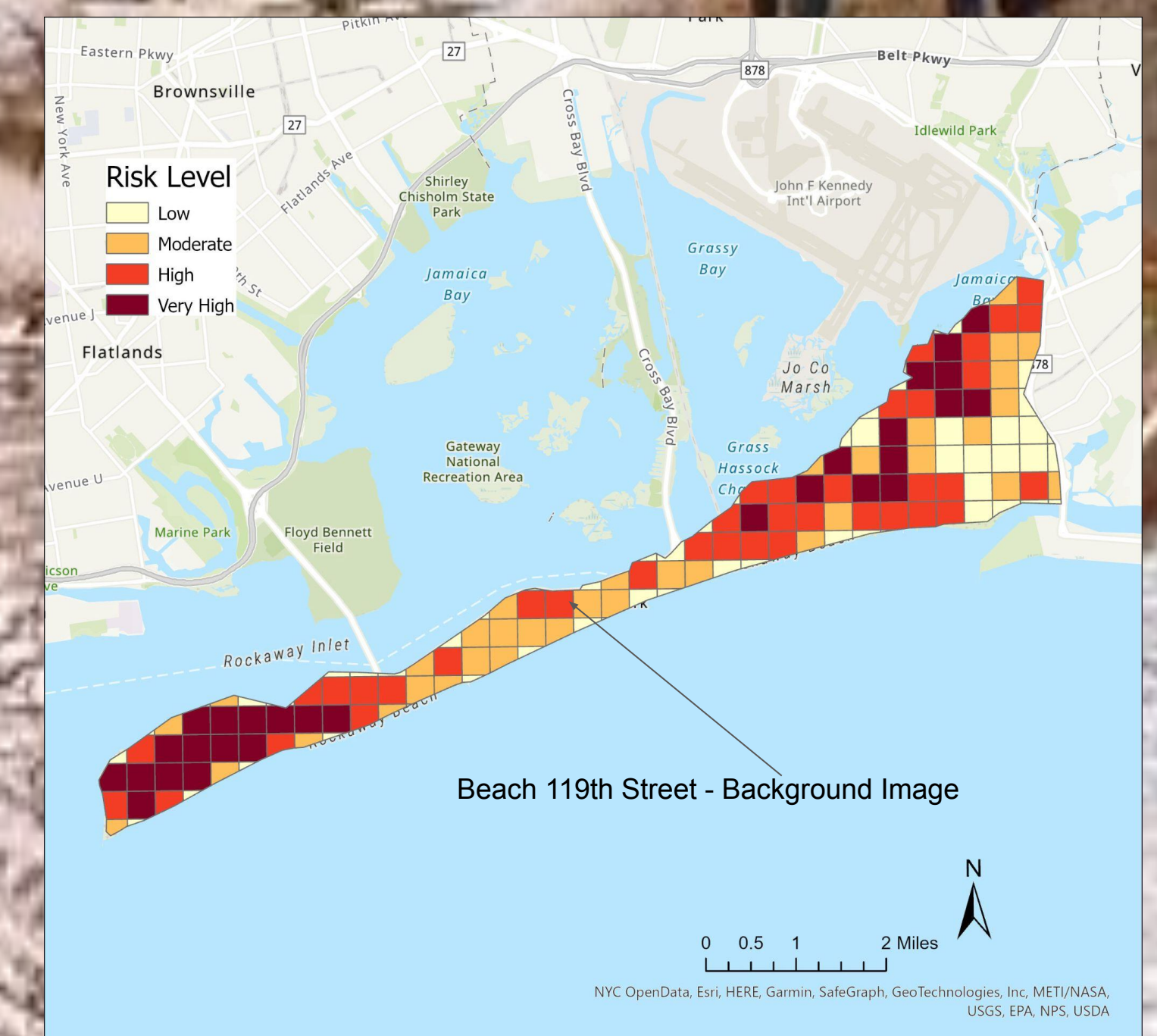


Figure 4: A classification GIS map that displays the risk level per cell of the grid based on flood rate per area. The risk levels were categorized into 4 levels ranging from low (yellow color) to very high (dark red color).

Discussion/Conclusion:

- Topography and dense population make flooding events more devastating
- High risk areas located in southwest and parts of northeast
- An area with a low topographic relief will flood more quickly causing a higher flood rate
- A high topographic relief area is flooded at a much slower rate due to higher elevations
 - Higher flood rate shows flooding will occur much faster with the same amount of water than for an area at a higher elevation
- Between zero and two meters of elevation the area of cell with highest flood rate by area will be completely flooded
- Lower risk of flooding as you move further inland
- First responders can prioritize high risk areas

Credit Authorship Contribution Statement:

Mead, N.: Figures, tables, GIS processing, ArcMap processing, data analysis in excel, poster, writing - Results, Discussion, Conclusion; Bourgeois, J.: editing, GIS processing, poster, writing - Abstract, Introduction, Methods, Discussion; Pelletier, A.: references, literature, ArcMap processing, editing, poster writing - Introduction; Badger, T.: editing, literature poster; Marsellos, A.E.: supervision, GIS processing, data analysis in excel, guidance, editing.

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